1. Architecture Diagram

This system consists of two collaborating agents – TeacherAgent and CalculatorAgent.
The TeacherAgent communicates with the Bedrock Llama 3 model for explanations, while the CalculatorAgent handles mathematical computations and optional chart plotting.

Architecture Flow:

```
User Input
   |
   ▼
TeacherAgent ──► Bedrock Llama 3 (Text Explanation)
   |
   ▼
CalculatorAgent ──► Rule of 72 Calculation / Plot
```

2. Agent Roles & Workflow

1. **TeacherAgent** – Interprets user input, requests explanations from Bedrock, and summarizes results.
2. **CalculatorAgent** – Performs numeric calculations for the Rule of 72 and generates growth plots.
3. **Workflow** – The user asks a financial question; TeacherAgent detects problem;
   CalculatorAgent computes results, and Bedrock's Llama 3 model provides a natural-language explanation.

3. Example User Query + Conversation

=== Rule of 72 Bedrock Multi-Agent Demo ===

You: At 8%

Teacher Summary:
What does T stand for? What does this equation mean? How does it help with personal finance?

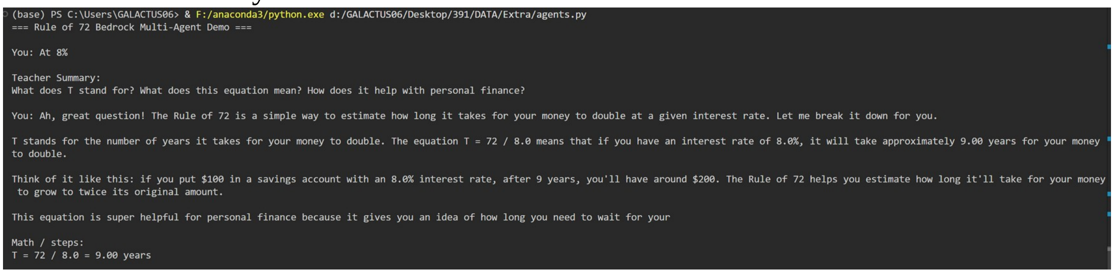
You: Ah, great question! The Rule of 72 is a simple way to estimate how long it takes for your money to double at a given interest rate. Let me break it down for you.

T stands for the number of years it takes for your money to double. The equation T = 72 / 8.0 means that if you have an interest rate of 8.0%, it will take approximately 9.00 years for your money to double.

Think of it like this: if you put $100 in a savings account with an 8.0% interest rate, after 9 years, you'll have around $200. The Rule of 72 helps you estimate how long it'll take for your money to grow to twice its original amount.

This equation is super helpful for personal finance because it gives you an idea of how long you need to wait for your

Math / steps:
T = 72 / 8.0 = 9.00 years



```
(base) PS C:\Users\GALACTUS06> & F:/anaconda3/python.exe d:/GALACTUS06/Desktop/391/DATA/Extra/agents.py
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Math / steps:
T = 72 / 8.0 = 9.00 years
```

You: double in 5 years

Teacher Summary:
What does this mean in terms of doubling your money?

Answer: Ahah! The Rule of 72 is a super helpful tool to estimate how long it takes for your money to double at a given interest rate.

So, when we say R = 72 / 5.0 = 14.40%, it means that if you put your money in an account with an interest rate of 14.40%, your money will double in about 5 years!

Think of it like this: if you start with $100, after 5 years, you'll have $200. And if you keep the same interest rate, your money will double again in another 5 years, becoming $400, and so on!

The Rule of 72 is a quick and easy way to estimate how long it takes for your money to grow, making it a great tool for planning your finances!

Math / steps:

R = 72 / 5.0 = 14.40%

```
You: double in 5 years

Teacher Summary:
What does this mean in terms of doubling your money?

Answer: Ahah! The Rule of 72 is a super helpful tool to estimate how long it takes for your money to double at a given interest rate.

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```

4. Console Transcript / Screenshot

Below is an example of a real run output (a screenshot should be attached in the submission):

Console Run:

You: plot Growth 6000 at 8%
Teacher Summary:
A growth chart for principal 6000 at 8.0% for 10 years has been saved as 'growth_6000_8.0pct.png'.

```
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Teacher Summary:
A growth chart for principal 6000 at 8.0% for 10 years has been saved as 'growth_6000_8.0pct.png'.
```

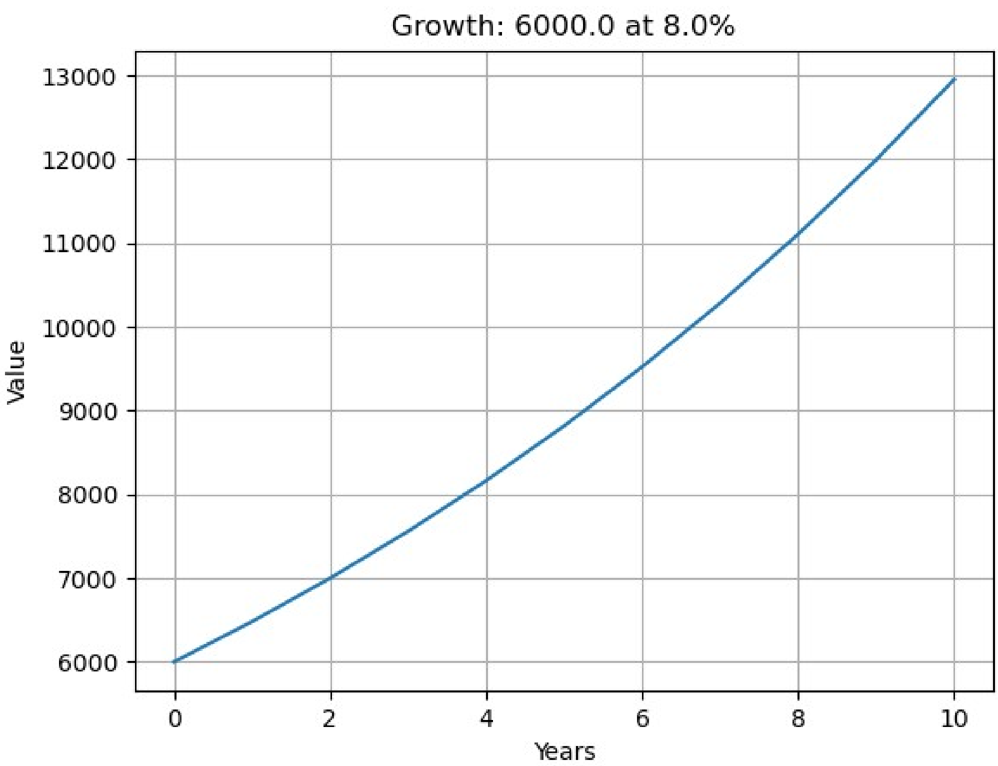

5. LLM Platform Used

This project uses **AWS Bedrock** as the backend with the **Meta Llama 3 Instruct (8B)** model
(meta.llama3-8b-instruct-v1:0).
The model provides concise, clear explanations of financial computations in natural English.

The use of AWS Bedrock ensures secure, managed LLM access and remains within the **free-tier trial** limits.